Agent-Based Infrastructures for Diachronic Digital Humanities

Cohort

Agents for retrieving, comparing and documenting textual evidence

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Acknowledgement

Thank you, Michael Radich

Texts

A modified, selected CBETA-derived corpus

Catalogue

Translator labels and mixed historical groups

Vocabulary

Curated strings for Dharmarakṣa research

Cohort builds the profiles and calculates the rankings.

Why we built Cohort

Let researchers direct agents and inspect the evidence behind their proposals.

Today's presentation

Use Buddhist texts to show the functions,
where results can mislead, and what the researcher decides.

We are demonstrating research tools, not reporting a new translator attribution.

Agent support for diachronic research

Study texts and their relationships across time.

Corpus retrieval

Find relevant texts and passages.

Semantic analysis

Find passages with related content.

Alignment

Locate matching wording and positions.

Interpretation

Propose explanations and alternatives.

Provenance Keep sources and checks attached to proposals.

Researcher decisions Inspect the evidence; accept or reject proposals.

Functions

Corpus	Find passages and read them in context.
Vocabulary comparison	Compare a text’s wording with texts labelled by translator.
Inquiry	Give agents a research question and inspect their actions.
Findings	Read the agents’ proposals, citations and checks.
Graph	Follow links between sources, proposals and decisions.

Example research questions

Two Lotus translations

Dharmarakṣa 竺法護 · T0263
Kumārajīva 鳩摩羅什 · T0262

Can we find related passages when the wording differs?

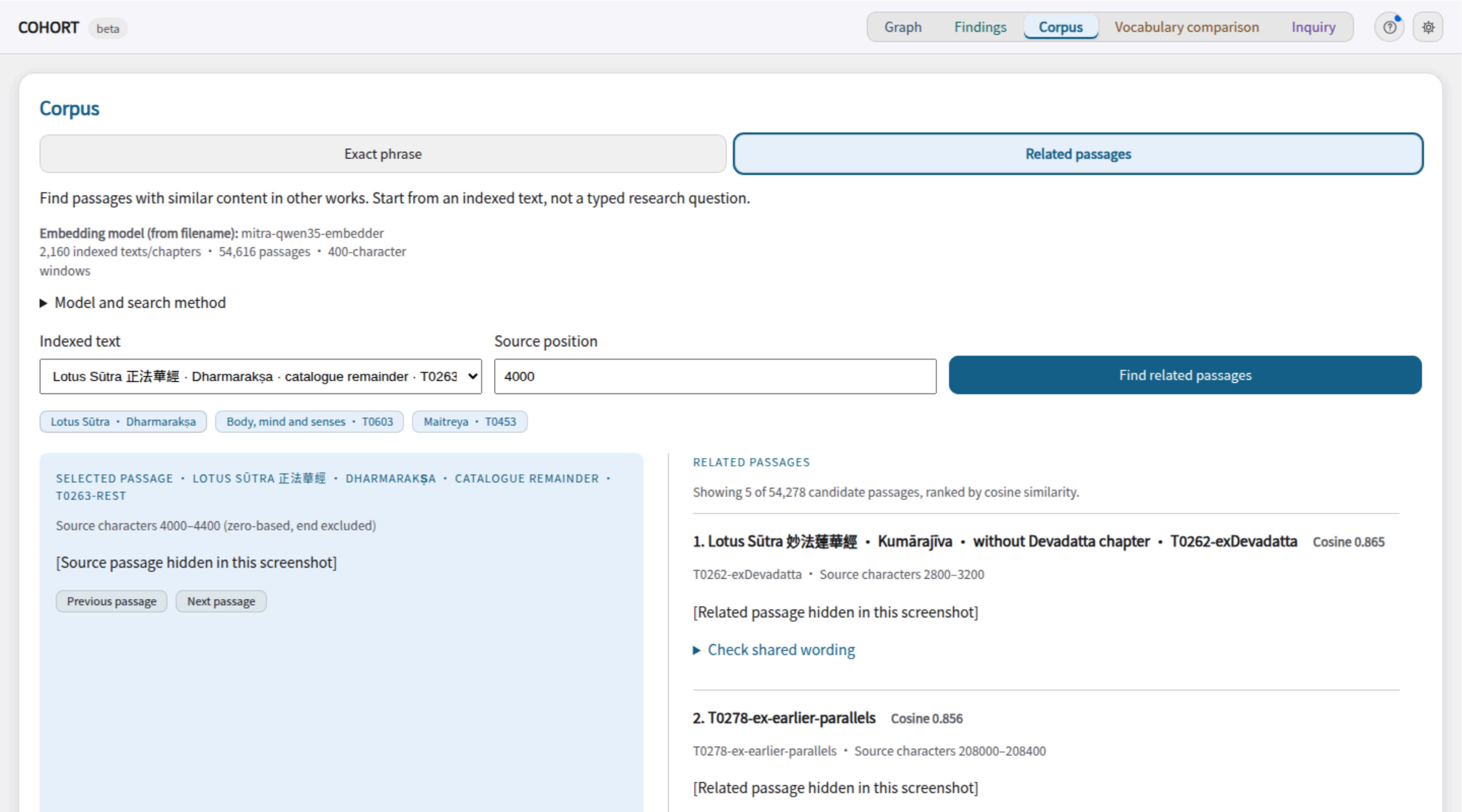
Text and later commentary

Yin chi ru jing 陰持入經 · T0603
Body, mind and senses

Could shared wording mislead us about who translated this text?

Known relationships let us inspect what the tools get right and wrong.

Find passages and read them in context.



Source passages hidden; controls and recorded outputs retained.

Find

Search exact wording or select Related passages.

Read

Selected passage on the left; candidates on the right.

Compare

Open shared wording and inspect source positions.

How related passages are found

mitra-qwen35-embedder

Selected passage

Use its stored vector

→

Cosine similarity

Compare with other works

→

Related passages

Rank candidates to read

Corpus → Related passages; agents can call semantic_neighbors.

Model name comes from the index filename; its exact revision is unrecorded.

Example: related content, little shared wording

Earlier Lotus translation

Dharmarakṣa · T0263-rest
Source position 4000

cosine 0.865

→

Later Lotus translation

Kumārajīva · T0262-exDevadatta
Source position 2800

Longest exact shared sequence: 3 Chinese characters

Shared wording is checked separately, with positions in both sources.

What vocabulary comparison calculates

Vocabulary list

Choose strings to count →

Target text

Count those strings ↓

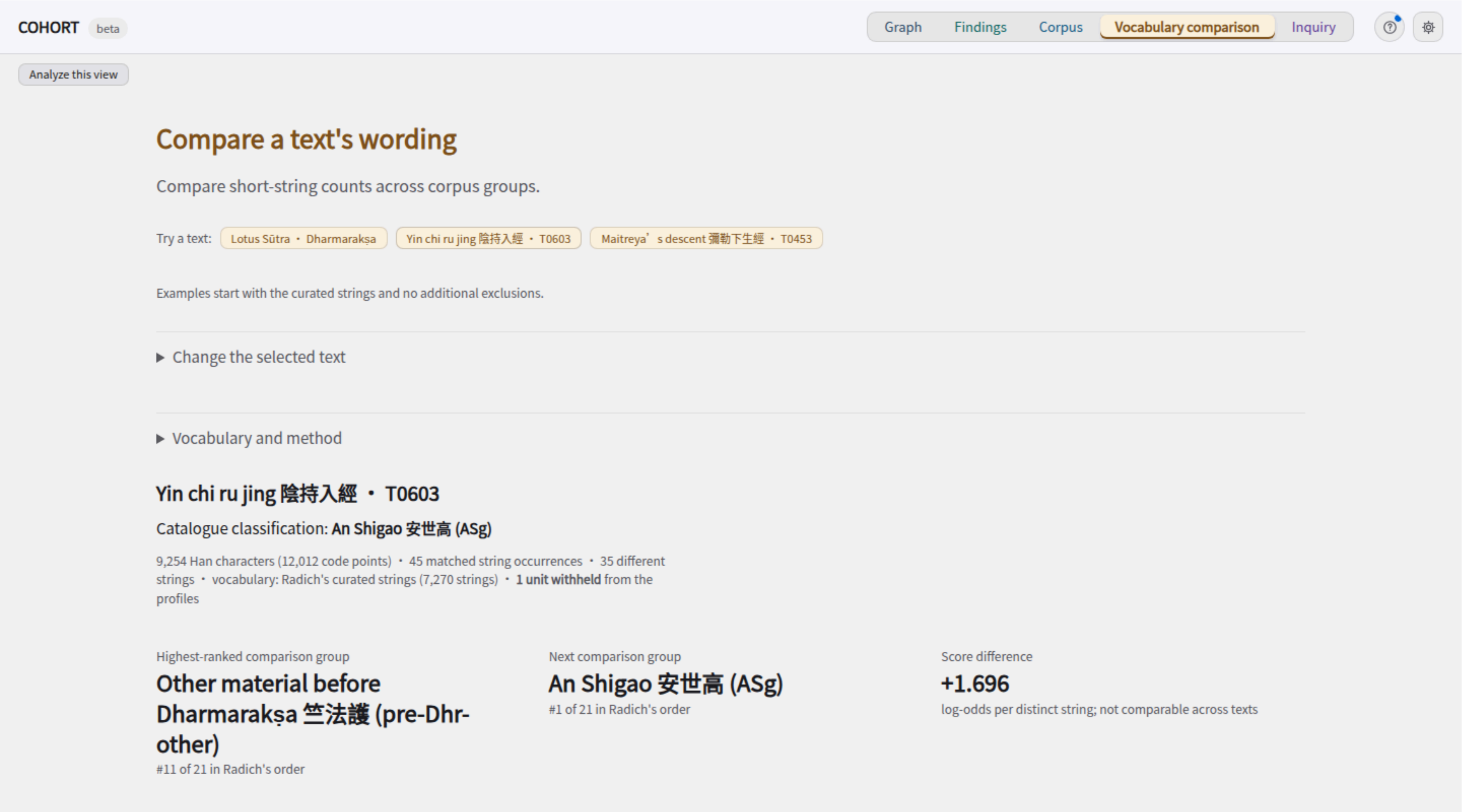
Group profiles

Texts pooled by catalogue label ↓

Compare usage → rank the catalogue groups

13 usable groups in this dataset. The target's whole work is withheld.

Compare a text’s wording with texts labelled by translator.



Source passages hidden; controls and recorded outputs retained.

Rankings

Which group’s string frequencies resemble this text?

Colours

Teal adds weight to A; rust adds weight to B.

Exclusions

Remove comparison material and calculate again.

T0603: remove the later commentary

With T1694

First: Other material before Dharmarakṣa 竺法護

Margin over next group: 1.696

→

Without T1694

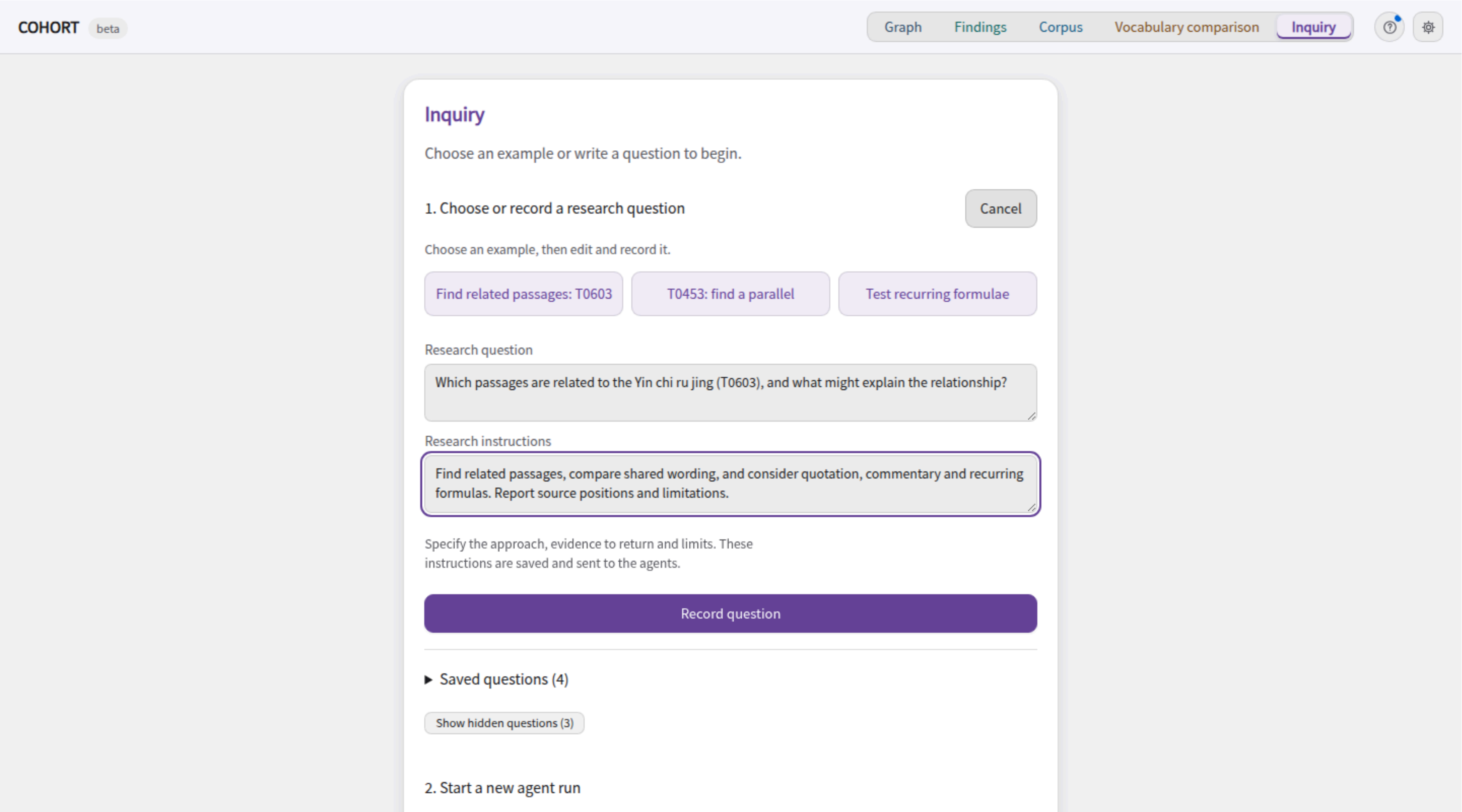
First: An Shigao 安世高

Margin over next group: 0.096

T1694 is the commentary on T0603. Its wording affects the ranking.

This agrees with the accepted attribution after exclusion; it does not establish one.

Give agents a research question and inspect their actions.



Source passages hidden; controls and recorded outputs retained.

Question

What do you want to find out?

Instructions

How should the worker investigate it?

Run results

Reopen the actions, inputs and outputs.

Worker, reviewer and analyst

Worker

Search and compare

Record proposals and citations

Reviewer

Different model family

Record citation checks and review verdicts

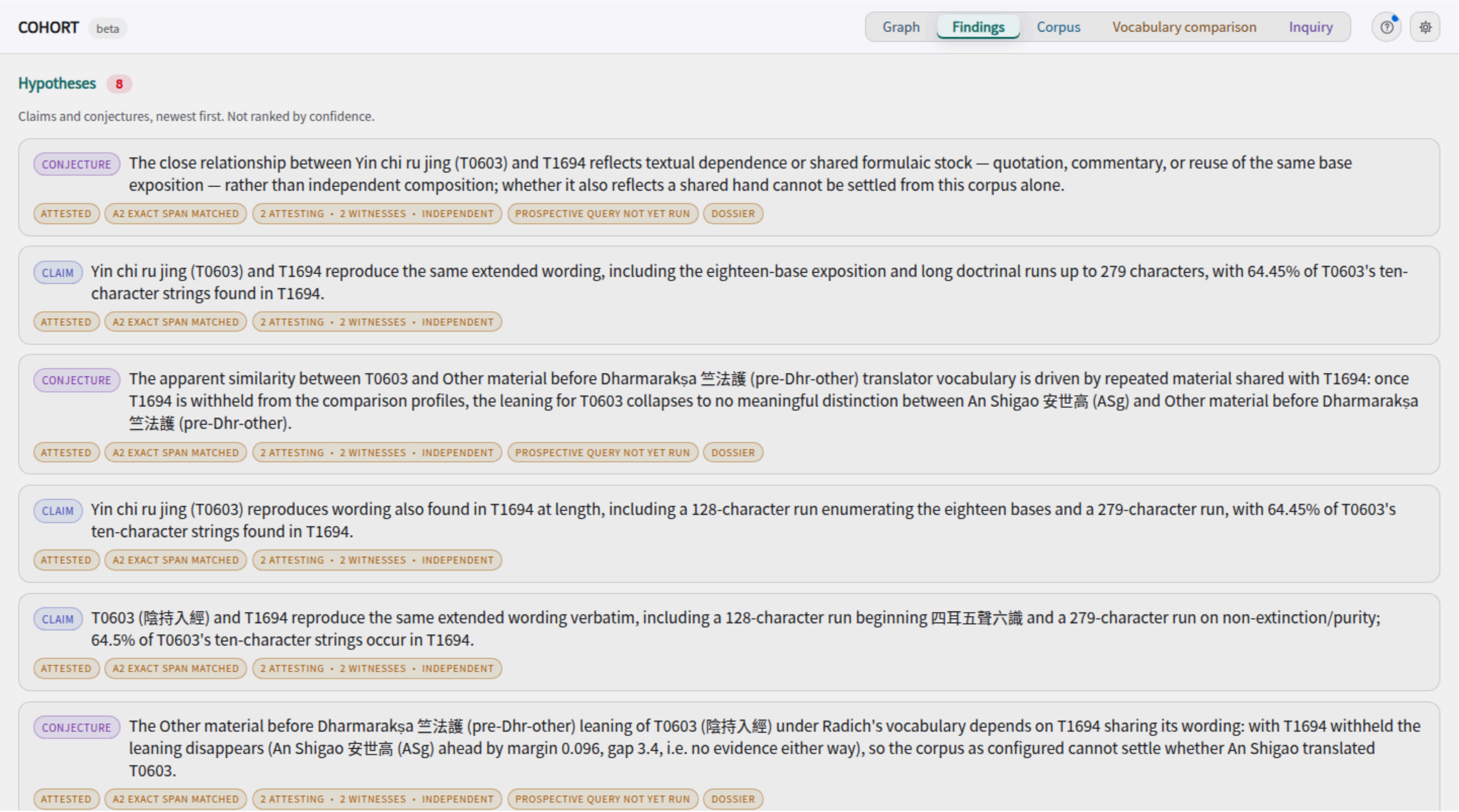
View analyst

Graph and vocabulary

Explain outputs; use read-only tools

Model names and actions are recorded. A reviewer can still be wrong.

Read the agents’ proposals, citations and checks.



Source passages hidden; controls and recorded outputs retained.

Proposal

A claim or a conjecture from the worker.

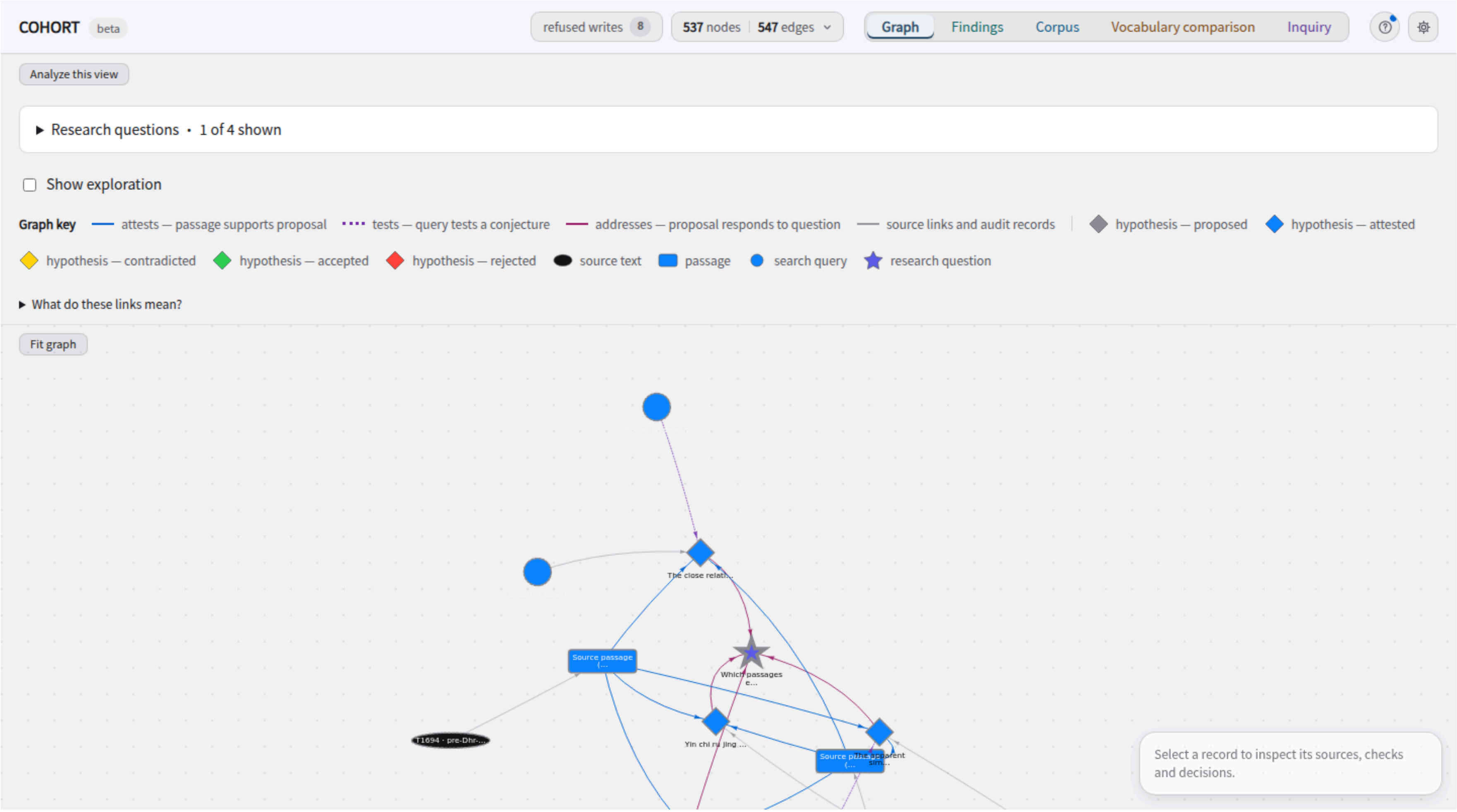
Sources

Read its citations and recorded checks.

Alternatives

Inspect what else could explain the observation.

Follow links between sources, proposals and decisions.



Source passages hidden; controls and recorded outputs retained.

Records

Sources, passages, proposals and questions.

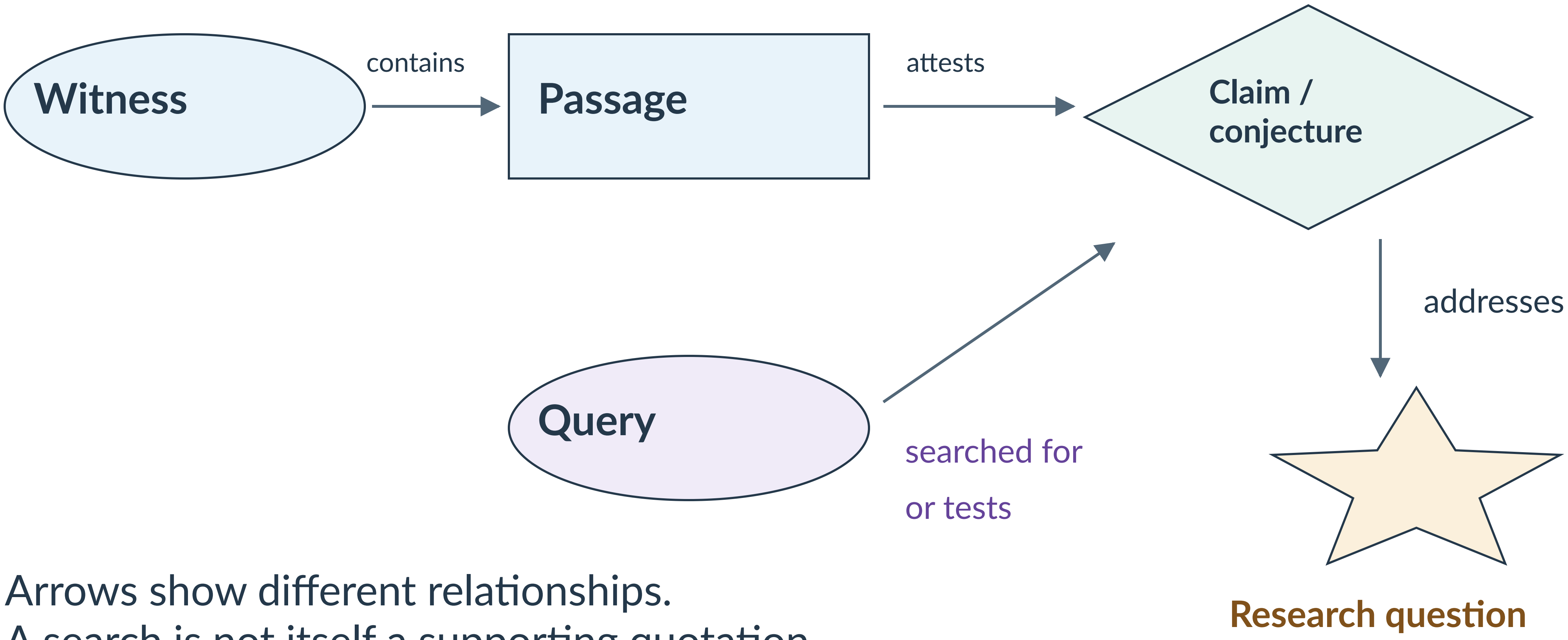
Links

Read the relationship label on each arrow.

Exploration

Show recorded tool activity beyond cited evidence.

Why a query points to a proposal



Arrows show different relationships.
A search is not itself a supporting quotation.

Checks and researcher decisions

Span verified

The quotation resolves at its source position.

Review recorded

Inspect the reviewer’s verdict and evidence.

Researcher decides

Accept or reject an eligible record.

Accepted passage ≠ accepted interpretation

Decisions stay in the record. The source text is unchanged.

Cohort

1	**Find**	Lotus passages and shared wording
2	**Compare**	T0603 with and without its commentary
3	**Inspect**	Saved actions, proposal, citations and checks

Results and limits

Useful outputs

Lotus: a related passage with little exact shared wording.

T0603: exclusion exposes the commentary's effect on the ranking.

Proposals, sources and checks can be inspected together.

Observed limits

Kumārajīva's Lotus wrongly ranks Dharmakṣema first in vocabulary.

One known-answer example cannot validate attribution.

Weak tests and reviewer errors still require researcher scrutiny.